

Week 6 Worksheet 1

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Exercise 1. Consider a spin-1/2 system.

- a) Find the eigenvalues and eigenspinors of S_z .
- b) Do the same for S_x and S_y , and write them in terms of $|+\rangle$ and $|-\rangle$, the eigenspinors of S_z .
- c) Consider the raising and lowering operators $S_{\pm} = S_x \pm iS_y$. Act with them on $|+\rangle$ and $|-\rangle$. How many states do you obtain?
- a) Since $S_z = \frac{\hbar}{2}\sigma_z$ and $\sigma_z = \text{diag}(1, -1)$ is diagonal, we can read off the eigenvectors as $(1, 0)$ and $(0, 1)$ with eigenvalues 1 and -1 , respectively.
- b) The part for S_x is done in Griffiths. For S_y , we calculate

$$\begin{vmatrix} -\lambda & -i \\ i & \lambda \end{vmatrix} = \lambda^2 - 1,$$

so that $\lambda_{\pm} = \pm 1$. This means the eigenvalues of S_y are $\frac{\hbar}{2}\lambda_{\pm}$. For the eigenvectors, we first check for $\hbar/2$ that

$$\begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix} \begin{bmatrix} \alpha \\ \beta \end{bmatrix} = \begin{bmatrix} \alpha \\ \beta \end{bmatrix}$$

has the solution

$$\alpha = -i\beta.$$

This gives an eigenvector

$$\begin{aligned} |+_y\rangle &= \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ i \end{bmatrix} \\ &= \frac{1}{\sqrt{2}}(|+\rangle + i|-\rangle). \end{aligned}$$

A similar calculation for $-\hbar/2$ gives

$$\begin{aligned} |{-}_y\rangle &= \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ -i \end{bmatrix} \\ &= \frac{1}{\sqrt{2}}(|{+}\rangle - i |{-}\rangle) \end{aligned}$$

c) You can calculate that

$$S_{\pm} |\pm\rangle = 0$$

while

$$S_{\pm} |\mp\rangle = \hbar |\pm\rangle .$$

So we get just the two states, and $S_{\pm}$ interchanges the up and down spin states (up to a factor of $\hbar$).

Exercise 2. A beam of spin-1/2 particles moving along the y -axis goes through two collinear Stern-Gerlach apparatuses, both with lower beams blocked. The first has its $\mathbf{B}$ field along the z -axis and the second has its $\mathbf{B}$ field along the x -axis (i.e., is obtained by rotating the first by an angle $\pi/2$ about the y -axis).

- a) What fraction of particles leaving the first will exit the second?
 - b) If a third filter that transmits only spin up along the z -axis is introduced, what fraction of particles leaving the first will exit the third?
 - c) If the middle filter transmits both spins up and down (no blocking) the x -axis, but the last one transmits only spin down the z -axis, what fraction of particles leaving the first will leave the last?
- a) 1/2, since if the particles have spin up along z that says nothing about their spin along x .
 - b) 1/4, since after passing through the x Stern-Gerlach apparatus, the particles have their spins projected onto the x -axis, which means their spins along z are now evenly spread out (no longer just positive!).
 - c) 1/2, since the splitting caused by the middle apparatus still changes the spin states of the particles leaving the first apparatus.